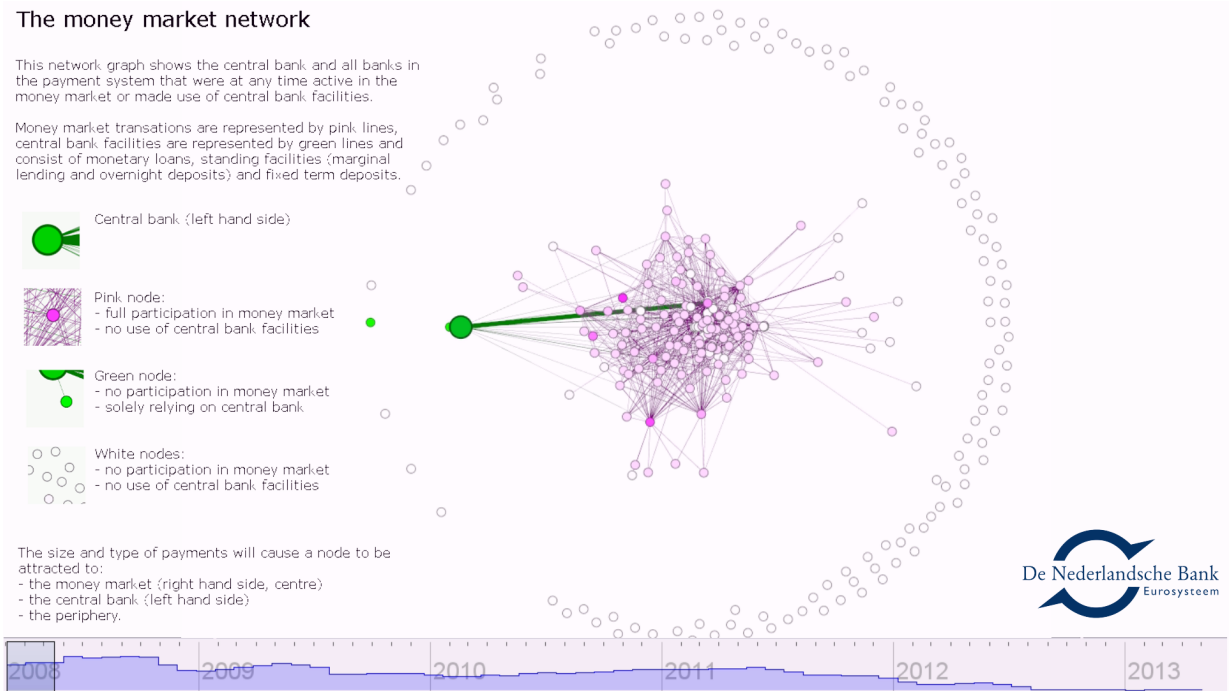
Dynamic visualization of large transaction networks: the Dutch overnight money market
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This paper shows how big data can be visualized in a dynamic way using case studies based on interbank payment transactions. First, we show how the interbank payment system could be monitored by looking at typical payment patterns over several days or, by collapsing the sample, for an average day.
Second, we zoom in on an important subset of interbank payments: unsecured interbank money market transactions, using a long time series from 1999 to 2013.

This paper aims to demonstrate 1) the feasibility of dynamic visualization of large data sets and 2) the potential gains from dynamic visualizations in terms of both analysis and presentation. We show that a regular laptop computer can be used to represent large data sets without aggregating these data. In addition, it is possible to create visualizations of large dynamic networks in a relatively routine manner. By virtue of recent developments in software engineering, the need for programming expertise to create network visualizations has been reduced considerably. While scripting remains useful to transform tabular data into a network structure, visualizations can be created with point and click software. As a result the visual approach to network analysis can become part of the regular analysts' toolkit.

The advantages of visualizations of big data as provided in this paper are twofold. First, all individual transactions between banks (or in general interactions between nodes) can be visualized. This means no initial loss of information as a result of aggregation at the start of the analysis. In payment systems analyzes daily aggregates are often used whereas in bank supervision monthly or even quarterly data are generally used. The obvious disadvantage of transaction data is that the visualization may become too noisy to identify anything meaningful. We countered this disadvantage by using different time windows to smooth excessive volatility. Second, our approach enables multiple dimensions to be included in one visualization.

The example video below shows the network of Dutch overnight money market transactions. In order to provide a more complete picture of the system's liquidity as a whole, all liquidity transactions to and from the central bank have also been included.



To start the video, please click on the figure.

To access the full paper please use this link.